Big Data and Machine Learning Methods

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Abstract

This chapter delineates big data and machine learning (ML) methods in innovation research. Large-scale data and computational methods are currently being used to advance innovation research on sources and processes. The chapter covers three oft-used computational methods: topic modeling for identifying latent themes, word embeddings for assessing semantic linkages, and generative large language models for text annotation and coding. Combined with large-scale data sets, these methods allow for nuanced and systematic analysis of innovation processes. The chapter also addresses the limitations of employing big data and ML methods in empirical studies, such as data quality and ethical considerations. It emphasizes the importance of human expertise and the potential of ML to aid as research assistants and interlocutors in social science innovation research.

Keywords

Big data · Innovation · Topic modeling · Word embedding · LLMs · Data quality

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1 Introduction

This chapter addresses the evolving landscape of innovation studies, driven by the increasing availability of big data and the application of machine learning (ML) methods. The emergence of large-scale digitized and "born-digital" data sources has opened new avenues for examining enduring questions in innovation studies, such as how newness emerges. This abundance of textual data, alongside advancements in computational analysis, allows researchers to study innovation processes with unprecedented breadth and depth. Foundational principles of innovation, such as recombination of existing elements, the temporal unfolding of ideas, processes of recognition, the interpretation of newness by diverse actors, and the categorization of new concepts, are now being investigated using computational tools. This shift enables a more nuanced understanding of the complexities inherent in innovation processes, which are often nonlinear and involve uncertainties and diverse interpretations.

First, the chapter introduces its notions of innovation and discusses challenges and advantages of old and new types of large-scale data. It proceeds to focus on the application of specific computational methods to the study of innovation: it introduces and discusses topic modeling, word embeddings, and generative language models as key analytical tools that leverage big data to provide new insights into innovation processes. These methods offer distinct approaches to analyzing large textual corpora, ranging from identifying latent thematic structures to mapping semantic relationships, and even generating text that mimics "understanding." By exploring the capabilities and limitations of these methods, this chapter is aimed at delineating how they contribute to and transform the study of innovation.

2 Innovation

Research on innovation studies innovative breakthroughs as they are the foundation for change in scientific and technological ideas, and potential increases in social and economic value. The focus of innovation research includes the sources and the effects of such breakthroughs. Research also addresses how organizations can innovate and, for example, change their business models.

This chapter focuses on the study of sources of innovation, for which research has identified several foundational principles and mechanisms. Innovations are based on some sort of recombination of actors, objects, or knowledge; new ideas emerge over time, and novel ideas must be recognized (for an overview see, for example, Cattani et al. 2023).

New actors, organizational forms, and objects, collectively referred to as newness, do not arise spontaneously. Instead, newness results "from combinations and permutations of what was there before. Transformations are what make them novel" (Padgett and Powell 2012, p. 2). Structurally, innovations emerge from the recombination of people, practices, and relations across multiple social network domains and "improve on existing ways of doing things" (Padgett and Powell 2012, p. 5) or

even "change the ways things are done" (Padgett and Powell 2012, p. 5). Cognitively, *recombination* of different kinds of knowledge—deep, local, and bridging—is necessary to create breakthroughs (de Vaan et al. 2015).

Other works in organizational studies have picked up on the complexities of innovation processes as journeys *over time and organizational spaces* (e.g., Garud et al. 2015). These studies examine innovation processes by observing the sequences of events that unfold over time, "events that unfold as ideas emerge, are developed, and are implemented within firms, across multi-party networks and within communities" (Garud et al. 2013, p. 776). Innovation processes are often not linear and may entail failures, so an empirical perspective over time as the process evolves is necessary.

Research also shows that to cope with uncertainties in innovation processes, *interpretations* of diverse sets of actors are involved. For example, in the history of biotechnology, entrepreneurs contested interpretations of "what biotechnology meant at any one point" (Kaplan and Murray 2010, p. 108) and constructed different economic logics of the field in which they operated. Similarly, industry analysts contested what a new breast cancer treatment was a case of (Mützel 2022).

The concept of "reflexive cognition" (Stark 2009, p. 4) analytically breaks down such interpretive processes, in which actors collaboratively create meaning, into three moments. First, actors must be able to *recognize and identify* something, such as a new product or practice, by making previously uncommon associations across various networks or knowledge domains. Second, the new thing requires *a label and a category* to belong to, and to which others may attach their interpretation and meaning. And third, *others* need to be convinced of the new thing so that they can *value* and support it and its *categorization*.

Categories seemingly reduce the cognitive complexities of social reality and provide a "conceptual system" (Rosa et al. 1999). Meaningful categories only stick over time when a first label is adopted and used by different actors (Grodal et al. 2015). Labels may stem from attempts to self-categorize (e.g., Vergne 2012), although such labeling is often contested (e.g., Slavich et al. 2020) as it needs to resonate with audiences or other intermediaries. Moreover, labels may serve as supportive vocabularies in communicating a new category (e.g., Hsu and Grodal 2021). In particular, new combinations of words and concepts help to signal a new category while also allowing for common ground to understand the meaning of that new category.

To coordinate amid uncertainty and ambiguity, actors use stories to temporarily settle on categories about what the situation is about. "Category emergence" and "category creation" are two possible avenues through which category formation can take place (Durand and Khaire 2017). In the process of "category creation," actors—in mostly cognitive moves—redefine and reinterpret preexisting attributes and features to elaborate how a new category is different from and normatively superior to an existing category (e.g., Rao et al. 2003). Instead, processes of "category emergence" involve labeling after a material innovation has been identified by actors involved in the market. In category emergence, the category acquires its legitimacy through the explication of the meaning of the category—most notably from third-

party intermediaries in discursive contestations (e.g., Grodal and Kahl 2017; Mützel 2022).

These principles of recombination, processes over time, cognition and interpretation, and processes of categorization and institutional logics are expressed in stories on innovations.

3 New and Old Large-Scale Data

As this brief review of innovation research has shown, actors involved in innovative processes talk and dispute over meaning and interpretation, leave written traces of such meaning-making processes, e.g., in meeting minutes and other documents, and can be observed and interviewed about how innovative breakthroughs failed or came about. Involved actors file patents, publish academic papers, and third-party observers and analysts produce special reports and articles on how to evaluate the claimed innovative breakthroughs. Accordingly, studies of innovation processes have employed various qualitative research methods to gather data on how novel ideas emerge, including ethnographic fieldwork, interviews, and document analysis.

For instance, patent data have long been used to study the recombination of ideas and their potential economic value using citation counts to patents (e.g., Griliches 1990). Researchers have followed innovation journeys over time based on ethnographic field work, interviews, and document analysis (e.g., Van de Ven et al. 1999).

Over the past two decades, libraries and archives have digitized old textual data sources, e.g., newspapers, organizational documents, and new "born-digital" data sources, e.g., online social media platforms, webpages, and digital-only text archives, have appeared (e.g., Gregoire et al. 2024). Characteristically, such data are large in size, unstructured, diverse—they include, for example, texts, images, administrative data, transactional data—and about an array of phenomena. Both old and new types of textual data have increased the variety and breadth of data sources used to study innovation processes—and they both come with caveats.

Trace data sourced from social networking sites have their own advantages and challenges. Such big data on communications are exciting for research because, contrary to scarce and expensive data sources of surveys and interviews, they are large, always on, and nonreactive. However, they are collected and further used by social media platforms for their profits (e.g., Zuboff 2019). Research must carefully consider the repurposing of such data, especially since they are incomplete and algorithmically confounded (Salganik 2017). Moreover, access to proprietary data from social media platforms is limited; indeed, at the time of writing, most social media platforms cannot or can no longer be accessed or web scraped for research.

Moreover, both digitized and digital data need further processing and curation before they can be analyzed. The transformation from analog documents to machine-readable text data takes several steps. It includes computer-assisted digitalization using algorithms such as optical character recognition or manual digitalization, transcribing the source material. The resulting machine-readable documents are rarely ready to use but must be further processed and curated as they may contain

errors or superfluous information. Whereas collections of digital text data allow for swift downloads of machine-readable files, they too need further curation and processing, such as nontextual information.

The growing literature on data quality (e.g., Hurtado Bodell et al. 2022) and pre-processing of unstructured, textual data (e.g., Hickman et al. 2022) points to the many consequential decisions during data curation that affect later data analyses and the validity of insights. Additionally, the selection of datasets on which algorithmic procedures are trained proves consequential as they may propagate unwanted yet encoded societal biases, stereotypical associations, and negative sentiment toward specific groups. To mitigate such effects and to increase transparency and accountability, researchers in academia and industry have emphasized the need to document the characteristics of datasets used in training and analyses (e.g., Whittaker et al. 2018). They suggest providing each dataset with information on its motivation, composition, collection process, and recommended uses (e.g., Gebru et al. 2021).

4 Methods of Data Analysis

Before looking in detail at current methods, it is helpful to briefly point to earlier methods of textual, network, and other relational analyses, as texts as data have long been used for social inquiries. For instance, using dictionaries and coding schemes, researchers have used content analysis to systematically find and measure specific constructs of interest (Krippendorff 2012). Word counts and frequency analysis in a given corpus have been used to describe and quantify "the manifest content of communication" (Berelson 1952, p. 18). Co-word analyses have helped to describe fields of knowledge (Callon et al. 1991).

Others have long been interested in identifying *latent patterns* in how individuals, groups, organizations, or fields make meaning. Many have studied discourses and narratives using qualitative analyses to trace meaning-making processes. Other research has employed formal methods that can capture the relationality of actors, events, objects, or terms, including network analysis, correspondence analysis, and Galois lattices, to detect how meaning is made. Such research has utilized various data types, including newspaper reports, directories, observations, and interviews.

Studies of innovation have been particularly interested in identifying latent patterns in texts of how innovative breakthroughs or novelty emerge. Research has used keywords, co-word analyses of keywords, and network analyses. For example, Sarah Kaplan et al. (2003) use word counts from letters to shareholders to measure firms' innovativeness. Mark Kennedy (2005) analyzes the co-mentioning of company names in newspaper data to analyze rivalries in the emerging market of computer workstations. Andrei Mogoutov et al. (2008) use a combination of publications, US patents, and abstracts of funded research projects to study the emergence of biomedical innovations. Using natural language processing (NLP), they extract semantically meaningful multi-term keywords from texts and plot their relations in semantic network analyses. Candace Jones et al. (2016) use network analysis of

actors' vocabulary to identify their institutional logics and how these logics shaped the creation of a new category called modern architecture.

The availability of large-scale digitized or digital data promises research to study processes with all their complexities and nuances over longer periods of time. Instead of scarce and expensively collected data points, digitized and digital data are now abundant. Instead of drawing samples, researchers can now work with "all the data" and obtain large-scale perspectives of the social.

No longer small or sampled, such large-scale data demand new methods of analysis. Research has pointed out that we are at "a watershed moment" (McFarland et al. 2016): The combination of available large-scale digitized corpora as well as novel tools from fields such as NLP and ML have elevated computational text analysis in empirical studies of the social world (for overviews, see, for example, Evans and Aceves 2016; Kobayashi et al. 2018; Borch and Pardo-Guerra 2025). Computational tools allow formally modeled insights into numerous social phenomena, events, situations, and relations by analyzing vast amounts of digitized text data. The interdisciplinary field of computational social science (e.g., Edelmann et al. 2020) has paved the way in how to combine powerful computation with careful social science and to propel automated text analysis (e.g., Grimmer et al. 2022).

The following sections introduce the current state of computational methods used on large-scale data sets in the study of innovation and discuss pertinent examples. Focus is placed on topic modeling, word embeddings, and generative language models. In this order, the methods also address the shift from models based on pattern recognition to those that can generate texts.

4.1 Topic Modeling

One of the methods that studies of innovation have used over the past decade is the unsupervised ML approach of topic modeling based on Latent Dirichlet Allocation, or LDA (e.g., Blei et al. 2003). Topic modeling yields groups of words that frequently co-occur and collectively represent *themes* or *latent topics*. This is based on a statistical model of language—that is, on a probabilistic model of how words are distributed in a corpus, which does not require a priori dictionaries or coding guidelines. Another advantage over hand-coding, frequency counts, and keyword analyses is that individual words may appear in different topics, reflecting their use in different contexts and various meanings. The amount of human involvement in topic modeling algorithms is minimal: researchers must tell the algorithm how many topics to find in the corpus; the algorithm then produces that number of topics, the words that make up each topic, and the distribution of those topics across the entire corpus.

The unsupervised method of topic modeling follows an inductive logic; it detects patterns in texts for explorative analyses—the idea is "to identify a number of substantively meaningful and analytically useful topics" (DiMaggio et al. 2013, p. 583). Fittingly, topic modeling has been likened to grounded theory approaches (Baumer et al. 2017), which identify themes based on close readings. Identified

topics present "the lens through which one can see the data more clearly" (DiMaggio et al. 2013, p. 582), whereas contextual knowledge is needed to interpret the patterns.

Several reviews and instructive discussions have introduced organizational scholars to the unsupervised ML method of topic modeling (e.g., Hannigan et al. 2019; Goldenstein and Poschmann 2019; Schmiedel et al. 2019). Sarah Kaplan and Keyvan Vakili (2015) make a compelling case for topic modeling as "a fruitful approach to measuring interpretations in the emergence of a new technological field" (p. 1441). They use topic modeling to measure the novelty of ideas as expressed in patents ("topic-originating patents"). Working with a large corpus of abstracts of US patents on nanotubes, they show the intricate relation of how innovative breakthroughs can but do not have to yield economic value. David Antons and Christoph Breidbach (2018) analyze the topical landscape in service innovation and service design based on scientific journal articles. Using topic modeling to identify topics of research, they map the connections between topics based on their level of distribution in articles as a topic network. Antons et al. (2019) use topic modeling together with rhetorical measures to study how the thematic substance and argumentative style of scientific articles impact their future use.

Another strand of research on innovation using topic modeling focuses on cultural dynamics, including categories, institutional logics, and styles. Analyzing restaurant reviews, Sophie Mützel (2015) shows the emergence and diminishing of culinary styles. Harsh Jha and Christine Beckman (2017) analyze the emergence of new organizational forms based on a corpus of charter school founding applications using topic modeling to show frames, organizational identities, and institutional logics. Richard Haans (2019) shows how distinctiveness affects performance and uses data from company websites to identify topics on how companies present themselves. Karl Taeuscher et al. (2021) also address the distinctiveness of entrepreneurial stories and market categories on a dataset of crowdfunding campaigns. In a study of the emergence of new breast cancer therapeutics, Mützel (2022) traces how science, industry analysts, companies, and journalists discuss and evaluate findings and failures of molecules and research strategies as topics. The analyses show the emergence of a new category ("targeted therapeutics") across the studied domains. Jessica Santana and Seonghoon Kim (2025) show how contested values shift and converge in codes of ethics in semantic network analysis and topic modeling.

Over the last decade, after the initial excitement, research has identified several problems with such unsupervised approaches. In general, unsupervised learning tools are data driven. This is useful for exploring a corpus, yet these tools cannot identify a priori specified theoretical concepts. Topic modeling is based on a statistical estimation of language and may not yield semantically meaningful topics. Indeed, as a generative model, it remains a black box with occasionally cryptic output. Moreover, topic models treat each document as a "bag of words," thus disregarding the syntax, grammar, or order of words within the document. Instead of statistical validity, research relies on semantic (are the topics produced semantically meaningful?) and external validity (do the topics and their distribution reflect external relevant events?). Overall, the unsupervised ML method of topic modeling

requires researchers to cope with "interpretive uncertainty" (DiMaggio 2015, p. 2). (Semi-supervised extensions of LDA, such as structural topic models (Roberts et al. 2019) and seeded topic models (Hurtado Bodell et al. 2024), have been developed to work with both the exploratory possibilities and the elements of focus of topic modeling.)

Laura Nelson et al. (2021) compare hand-coded, dictionary, off-the-shelf, supervised ML, and unsupervised topic modeling tools used to identify inequality in a corpus of articles. They find that topic models can complement traditional approaches to coding complex and multifaceted sociological concepts. However, they cannot fully replace traditional, labor-intensive approaches. The combination of widespread interest in topic models and their limitations paved the way for the adoption of other ML approaches.

4.2 Word Embeddings

Developed in the 2010s, word-embedding models (e.g., Word2Vec, Mikolov et al. 2013) are tools for reducing the dimensionality of text and extract numeric representations of concepts. Word-embedding models are unsupervised ML algorithms. They can be derived from any corpus and use a pretrained or a custom-trained model. Word embeddings create a vector representation of every word in a given corpus: they calculate the proximity and distance between terms and identify associations between these words and specific concepts. The idea is that the concept or meaning a word represents can be deduced by the distribution of words that surround it. The resulting semantic space thus synthesizes relationships between words, captures those relationships, and allows for various operations on words or concepts. Social science research views these spaces as "conceptual spaces" that are multidimensional and "within which concepts ranging from norms and knowledge to ideas and inventions relate to one another" (Aceves and Evans 2024, p. 789) and refers to word-embedding models as "a means of mapping meaning space" (Stoltz and Taylor 2021, p. 2).

To be sure, word embeddings reflect stereotypical racial, ethnic, and gender biases present in the texts on which they are trained (e.g., Caliskan et al. 2017). Some studies attempt to "de-bias" word embeddings (e.g., Bolukbasi et al. 2016), whereas others use these associations of biases to reflect the contours of cultural formations and trace changes in meaning over time (e.g., Nelson 2021).

Studies of innovation use word embeddings because of their capacity to efficiently extract words from a corpus that are semantically related. Jaewoo Jung et al. (2024) use word embeddings in conjunction with additional text analytic tools, first to extract terms and then topics from the letters of CEOs to shareholders to better understand what the notion "innovation" entails. Katharina Lix et al. (2022) evaluate the discursive breadth of team members working on different stages of software projects. The study tracks the variety of conceptual engagement at every stage of the project, demonstrating that high-performing teams can adjust their thinking to shifting tasks; these teams also exhibit a wider discursive breadth when generating

new ideas and a narrower breadth when transitioning to tasks that call for coordination. The research suggests that alternative text analytic methodologies would have found it challenging to capture such a nuanced concept of knowledge engagement. In a different setting, Bas Hofstra et al. (2020) analyze the innovativeness of scientific ideas. Using word embeddings from over one million dissertation abstracts, this study estimates the semantic location of concepts in a vast network of interrelated concepts and compares how newer concepts are positioned with regard to one another in that space.

Pedro Aceves and James Evans (2024) provide an instructive roadmap, an empirical example based on patent abstract data, and a systematic literature review on word-embedding models. Word embeddings are particularly interesting to scholars of innovation because they address the emergence of categories: "Categories are groups of entities with shared characteristics and attributes. As noted previously, concepts are the mental representations of categories" (p. 791). They show that word embeddings can "measure similarity within the conceptual space that existed at the time the patent was published" and contend that the theoretical notion and empirical application of conceptual space "has the potential to spur new theoretical developments and sharpen established theories" (p. 804).

Yet, like other text analytic methods, word embeddings have limitations (Rodriguez and Spirling 2022). The results of word embeddings may be inaccurate and, like other unsupervised methods, provide users with little control over what they study. Word embeddings cannot consider negation, polysemy of terms, syntax, or word order.

4.3 Outlook and Trends: Contextual Embeddings and Generative Large Language Models

A recent advancement in word-embedding models consists of contextual embeddings that rely on more complex, deep learning algorithms. They are built on transformer models (Vaswani et al. 2017) and trained on millions of words and thus part of the family of large language models (LLMs). Prominent transformer models are bi-directional encoder representations from transformers (BERTs) and generative pre-trained transformers (GPT). Both differ in their architectures, in how they specifically process and generate language, and how they are used in research.

Because of their bidirectional architecture, BERT models (Devlin et al. 2019) take into account the context of each occurrence of a word. Whereas previously mentioned embedding models are able to learn similar representations for the words that appear more frequently close to each other in a corpus and thus can assign semantically similar words a similar space, they are unable to capture the meaning of words in different contexts—e.g., "bank" may mean different things in different contexts. BERT can now capture the contextual variation of each word based on the sentence, paragraph, or document. This increases BERT's ability to generate language based on context by suggesting meaningful strings of words and sentences. BERT is publicly available, allowing users to train the neural network on new data and in

various languages. BERT models are trained on an extensive breadth of data, and users can plug in their data sets for analysis. BERT models are pretrained for various tasks, including text classification and sequence labeling. Moreover, BERT models can be fine-tuned for a particular task.

Research shows that trained BERT models are valuable tools for social science research: Salomé Do et al. (2022) show how the use of BERT models can help researchers in coding procedures, thus augmenting their capabilities. Moreover, fine-tuned BERT models can identify rhetorical styles (Bonikowski et al. 2022) and detect hate speech (Davidson 2025).

Studies in innovation research, including research on patents and categories, have also used BERT. Milan Miric et al. (2023) compare different text analytical approaches, including word embeddings and BERT, to identify patterns in US patents on artificial intelligence. Analyzing the "stickiness of category labels," Balázs Kovács et al. (2024) study genres based on books' synopses as found on the website goodreads.com. Using a BERT model and human labelers, they show constraints on category positioning in a market: label expectations stick from a prior to a next book publication, which may lead to label mismatches.

One extension of BERT models is BERTopic (Grootendorst 2022). This method combines sentence embeddings using BERT with a clustering of documents and yields a representation of topics. BERTopics presents topics as a distribution of words. This allows modeling of dynamic and evolutionary aspects of topics. To systematically review and sort the existing literature in the field of early innovation search that uses NLP approaches, Julian Just et al. (2024) run BERTopic. The analysis finds 18 distinctive innovation practices, such as "patent mapping" or "attribute recombination," based on textual analysis methods, such as topic modeling or text embeddings. Just et al. (2024) also use BERTopic to sort the results of a crowdsourced idea contest and to find patterns.

Generative language models of the GPT family of models (Ouyang et al. 2022) are also trained on vast corpora of texts and are already fine-tuned by human annotators (Mützel and Ollion 2025). In by now familiar conversational bots, such as ChatGPT, Claude, or Gemini, these models generate human-like responses to prompts, provide advice, deliver code, translate and summarize texts, generate texts and images, draft and improve texts, and collaborate across different programs. Technically, whereas BERT takes the context of each word to be generated into account, GPT models generate word by word based on the previous word it generated and the estimated probability of the next word to follow.

These conversational models are already "game-changers for science" (van Dis et al. 2023) and will also impact innovation studies: e.g., they can summarize, develop, evaluate, and translate knowledge (Grimes et al. 2023), formulate new propositions (Cornelissen et al. 2024), improve survey research, experiments, agent-based models, or vignettes with synthetic textual or image data (Bail 2024; Karell et al. 2025; Davidson 2024). At the same time, there are various limitations and potential dangers of these models, which can negatively impact research, including generated texts and sources that are factually inaccurate, exhibiting human and "machine biases" (Boelaert et al. 2025), are unethically sourced, cannot be

replicated, produce seemingly polished "junk science" (Bail 2024), and may in the medium and long term also negatively impact creativity (Hsu and Bechky 2024). Current research also points to the dangers of using closed, proprietary LLMs for research (Ollion et al. 2024) and instead suggests open-source LLMs (Palmer et al. 2024).

Given all these limitations, research shows that these models nevertheless open up new possibilities for researchers interested in analyzing large-scale text corpora. In this instance, GPT models are used as *virtual research assistants* that respond to natural language instructions and can eventually augment the capabilities of social scientists. Fabrizio Gilardi et al. (2023) demonstrate that using ChatGPT for annotations of Tweets and newspaper articles produces more accurate results than from hired workers who had received the same instructions. In their comprehensive discussion of classifying and coding texts in the social sciences, Caleb Ziems et al. (2024) show that instructed LLMs can reproduce human coding decisions. They recommend integrating LLMs in the research pipeline to transform large-scale data labeling. Yet, they also stress the need for human expert annotators who work in tandem with these models, instructing LLMs, proving hand coding only for refinement, and validating outputs.

The use of LLMs thus enables social scientists "to examine corpora of unprecedented size with unforeseen speed" (Bail 2024, p. 5). At the same time, using LLMs as interlocutors to interpret and categorize texts based on coding instructions delivered in prompts may also transform processes of qualitative text analysis. Research shows that LLMs "provide a productive blend of automation and nuanced understanding" (Than et al. 2025, p. 880).

Recent developments in text analytical models allow research interested in the study of innovation to work with large-scale text corpora in their analyses. All models discussed have advantages and caveats. Although topic modeling and word embeddings can show temporal dynamics, inductive mappings, and categorization processes, the capabilities of more recent LLMs suggest the possibility of detecting more nuanced processes of recombination and cognitive shifts.

5 Conclusion

This chapter has explored the burgeoning field of innovation research that uses big data and ML methodologies. It has shown how the increasing availability of digitized and born-digital data has provided new opportunities to study the intricate processes of innovation, from the recombination of ideas and knowledge to the temporal unfolding of innovation journeys and the construction of meaning and categories.

Computational methods such as topic modeling, word embeddings, and generative language models have significantly contributed to the field. Topic modeling offers an inductive approach to uncovering latent themes and patterns in large textual corpora, enabling researchers to explore the emergence of new technological fields and cultural dynamics. Word embeddings provide a means of mapping semantic

relationships between concepts, allowing for the analysis of conceptual spaces and the measurement of novelty. More recently, contextual embeddings and LLMs such as BERT and GPT models have demonstrated the ability to understand language in context and even generate human-like text, paving the way for advanced text analysis tasks and the potential for LLMs to serve as virtual research assistants and interlocutors. Moreover, these models allow for a renewed recombination of automated and qualitative approaches (e.g., Brandt and Timmermans 2021).

Despite the potential for novel insights outlined, it is crucial to acknowledge the limitations and challenges associated with these computational approaches. Unsupervised methods, such as topic modeling and word embeddings, can suffer from interpretive uncertainty and may not always correspond to a priori theoretical concepts. All ML models are data driven, and their outcomes can be influenced by data quality, pre-processing decisions, and potential biases encoded in the training data. Although generative language models are powerful, they raise concerns about factual accuracy, ethical sourcing, reproducibility, and the possibility of "junk science." As a result, when using these tools, a reflexive approach is required, emphasizing the ongoing need for human expertise to guide the analysis, interpret results, and validate findings.

Looking ahead, several trends can be noted. The use of contextual embeddings and generative language models promises nuanced and powerful analyses in the study of innovation. The ability of LLMs to perform tasks such as coding, summarizing, and formulating new propositions points to a future in which LLMs can significantly augment the capabilities of innovation researchers. The digitization of historical archives and the proliferation of born-digital data will further expand the scope and depth of inquiry in innovation studies. Analyses will also expand their data sources to include texts, sounds, and images as data.

In conclusion, the intersection of big data and ML fundamentally contributes to the study of innovation. These computational tools offer opportunities to investigate long-standing questions with new data and analytical rigor, revealing intricate patterns and dynamics in the emergence and evolution of newness. However, realizing the full potential of these possibilities requires a thoughtful engagement with their methodologies and limitations and a careful interplay between computational power and human expertise.

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